

Verifying Dynamical Systems: Stability

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Verifying cyberphysical systems

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Plan

- Dynamical system models
 - Stability
 - Lyapunov method
 - Stability of linear systems

Review dynamical systems

Physical processes are modeled by ordinary differential equations (ODEs) also known as dynamical systems

ODEs describe how variables change with time and inputs

$$\frac{dx(t)}{dt} = f(x(t), u(t)) \quad (1) \text{ where time } t \in \mathbb{R}; \text{ state } x(t) \in \mathbb{R}^n; \text{ input } u(t) \in \mathbb{R}^m; f: \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$$

In short this is written as $\dot{x} = f(x, u)$

Given an ODE (1) and initial state $x_0 \in \mathbb{R}^n$ and input $u: \mathbb{R} \rightarrow \mathbb{R}^m$, a behavior of the system is a state trajectory or *solution* $x: \mathbb{R} \rightarrow \mathbb{R}^n$.

Theorem. If $f(x(t), u(t))$ is Lipschitz continuous in the first argument and $u(t)$ is piecewise continuous then (1) has unique solutions.

Review. Linear dynamical system

$$\dot{x}(t) = A(t)x(t) + B(t)u(t)$$

Linear time invariant (LTI) system $\dot{x}(t) = Ax(t) + Bu(t)$

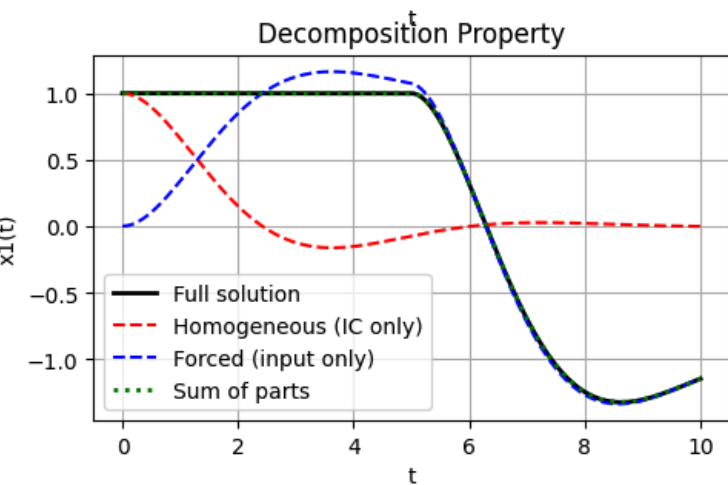
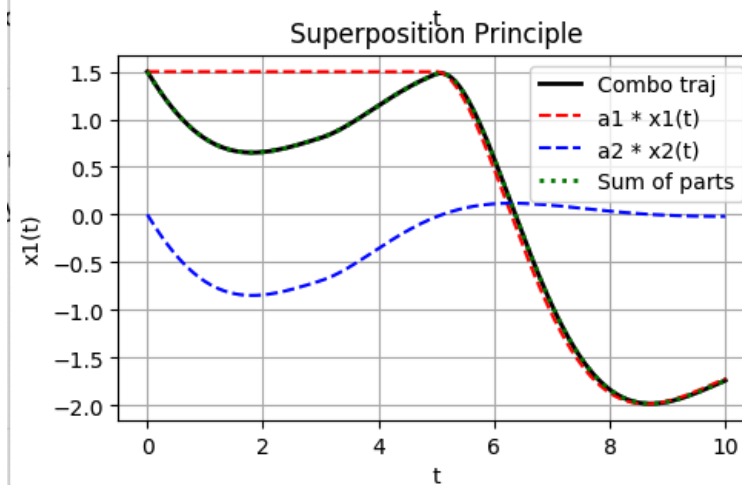
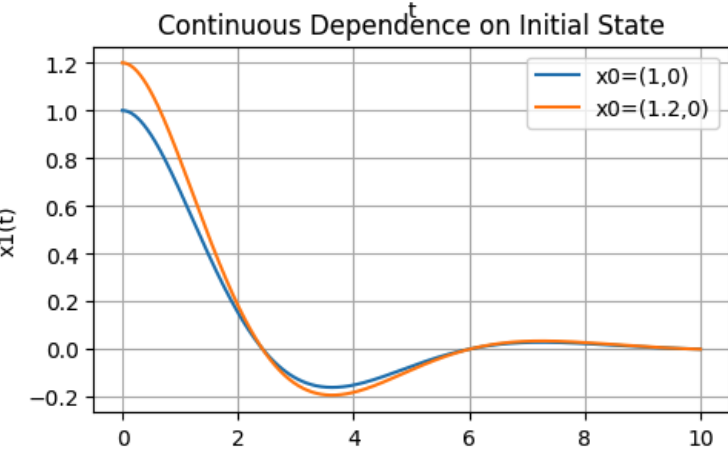
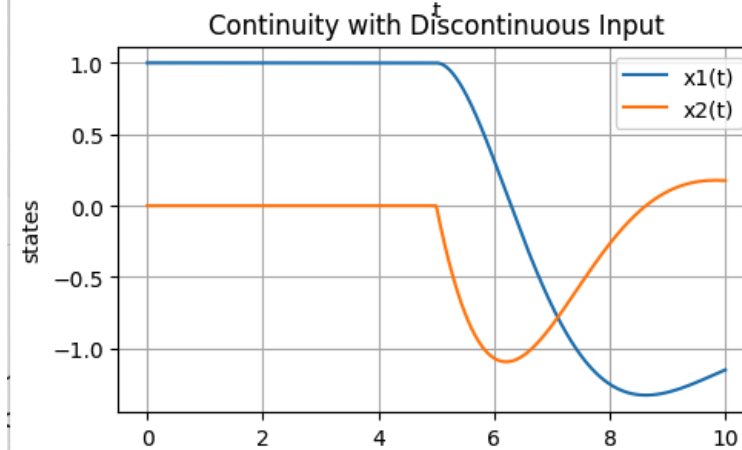
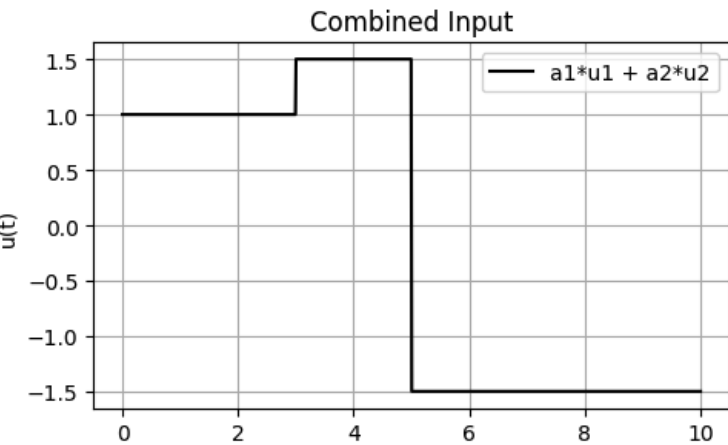
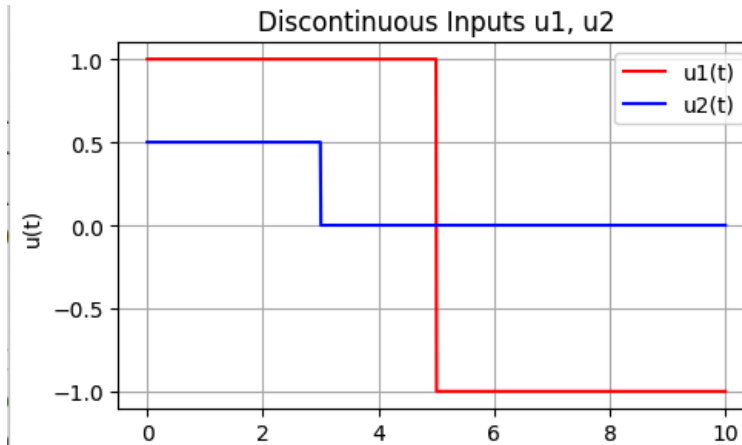
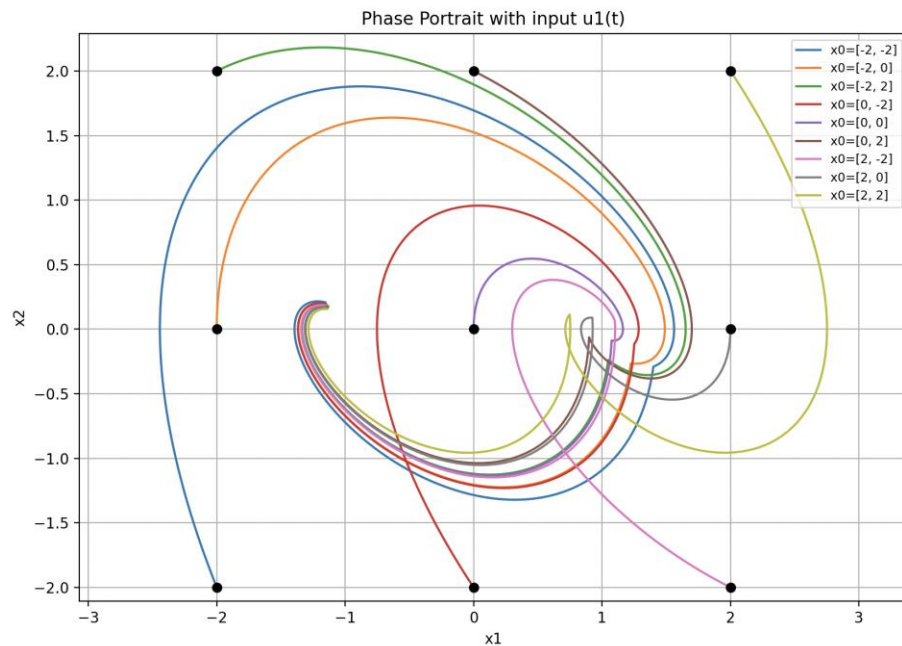
Theorem. Solutions of LTI system:

$$x(t, x_0, u) = x_0 e^{At} + \int_{t_0}^t e^{A(t-\tau)} Bu(\tau) d\tau$$

Properties of linear dynamical systems

$$\dot{x}(t) = A(t)x(t) + B(t)u(t)$$

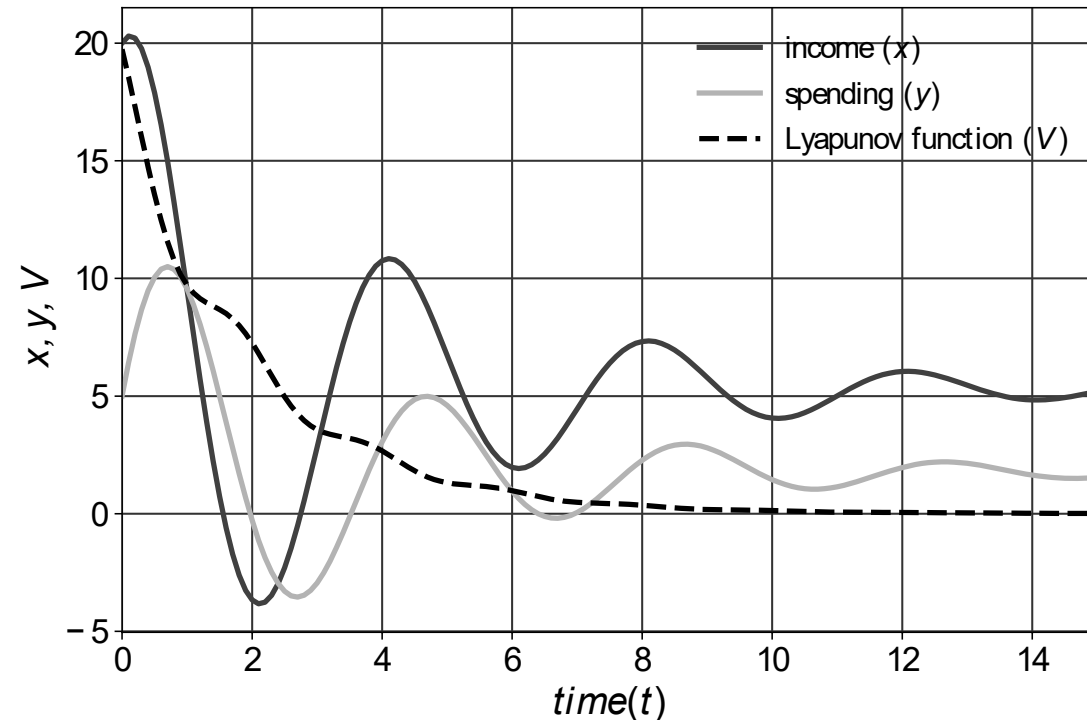
$$A = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



Example: Simple model of an economy

- x : national income
- y : rate of consumer spending
- g : rate government expenditure
- α : propensity to consume
- β : responsiveness of consumption

- $\dot{x} = x - \alpha y$
- $\dot{y} = \beta(x - y - g)$



Example: Gradient flow

- Consider the regression problem of fitting a straight line to data using gradient flow. The behavior of this learning algorithm can be modeled and analyzed as a dynamical system
- Given data: $\{(x_i, y_i)\}_{i=1}^n$ we want to fit a linear model $\hat{y}_i = \theta x_i$ by finding the **best** θ
- Minimizing the squared loss: $L(\theta) = \frac{1}{2} \sum_i (\theta x_i - y_i)^2$
- Gradient of loss $\nabla_{\theta} L(\theta) = \frac{1}{2} \cdot 2 \sum_i (\theta x_i - y_i) x_i = \sum_i (\theta x_i - y_i) x_i = \theta \sum_i x_i^2 - \sum_i x_i y_i = \theta A - B$
- Gradient descent $\theta_{k+1} = \theta_k - \eta \nabla_{\theta} L(\theta_k)$, η is a parameter called learning rate
 $= \theta_k - \eta(A\theta_k - B)$
- Gradient flow is the continuous-time limit

$$\frac{d\theta(t)}{dt} = -\nabla_{\theta} L(\theta(t))$$

$$= -A\theta + B$$

LTI system

- Solution $\theta(t) = \theta^* + (\theta(0) - \theta^*)e^{-At}$ $\theta^* = \frac{B}{A}$

$$f: \mathbb{R}^n \rightarrow \mathbb{R}$$

$$\text{Gradient } \nabla_x f = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right)$$

$$\text{Example: } f(x, y) = x^2 + y^2$$

$$\nabla_x f = (2x, 2y)$$

Gradient flow convergence to an optimal model

Consider the regression problem of fitting a straight line to data using gradient flow. The behavior of this learning algorithm can be modeled and analyzed as a dynamical system

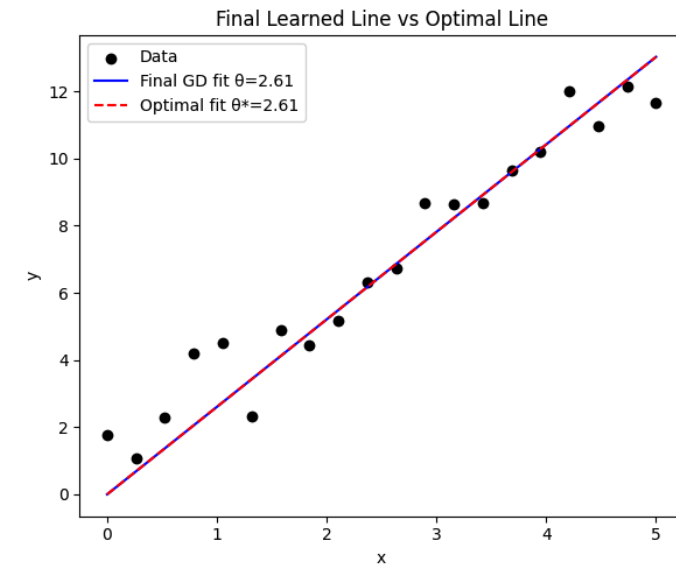
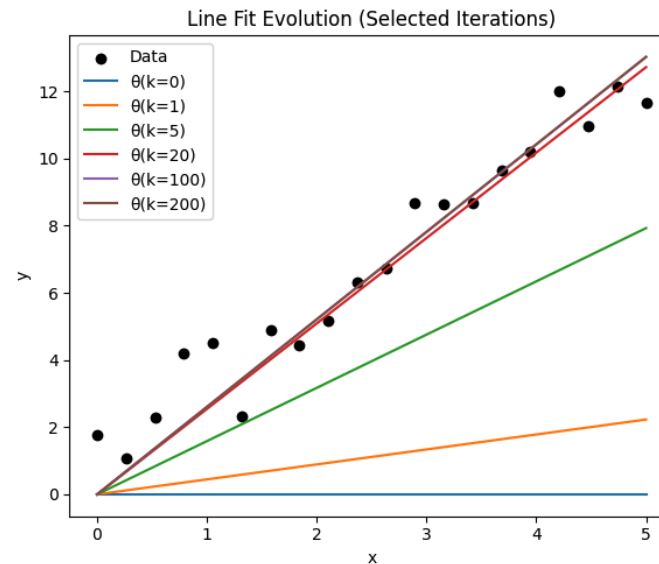
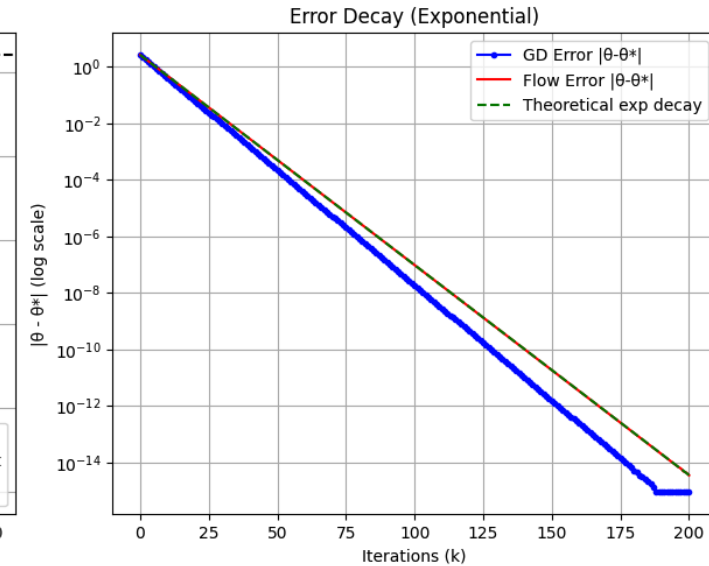
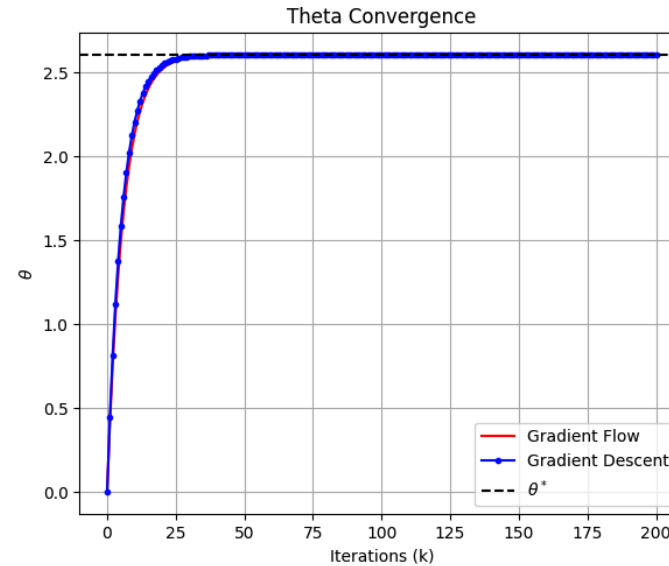
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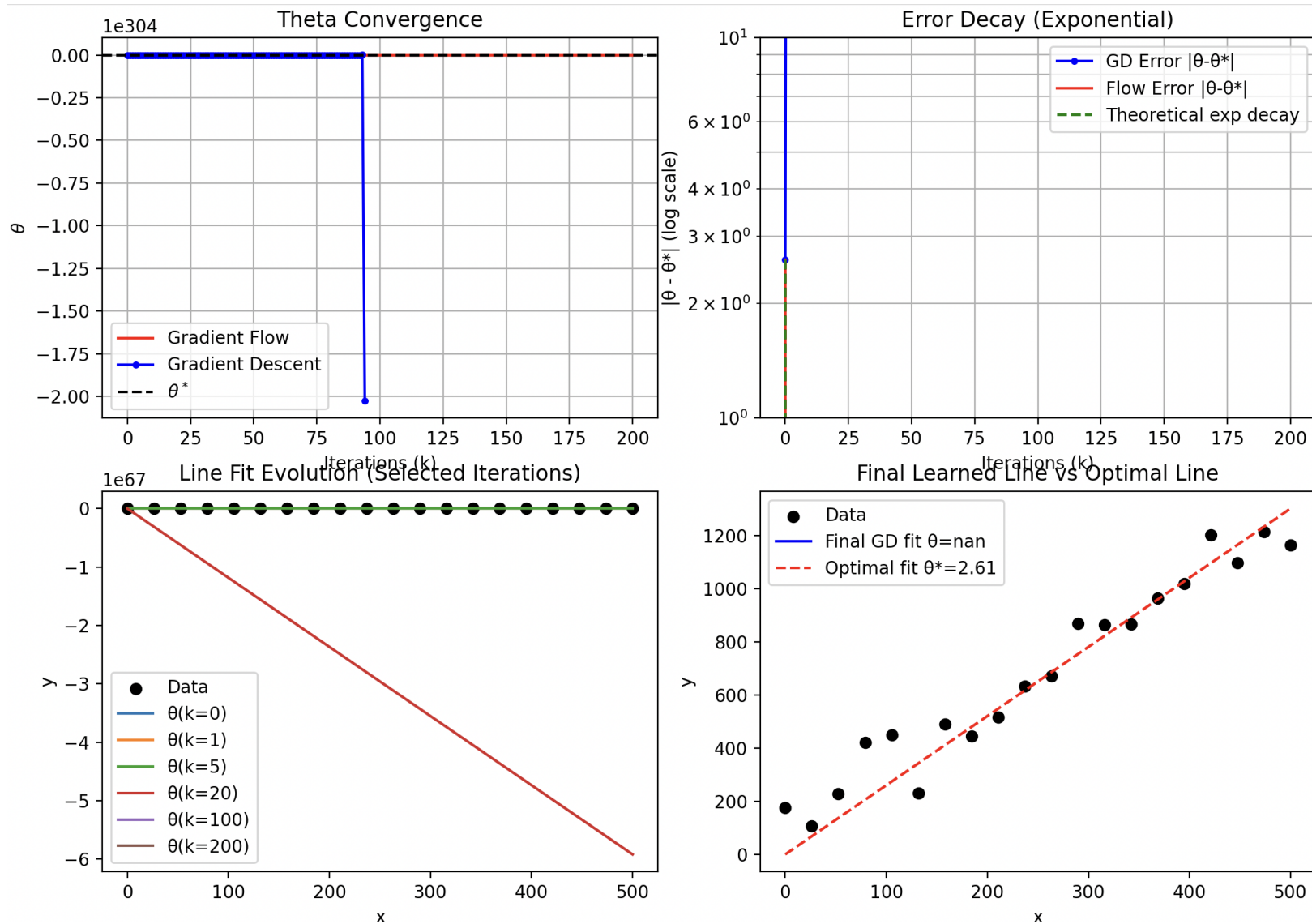
Gradient of loss $\nabla_{\theta} L(\theta) = \theta A - B$

Gradient descent $\theta_{k+1} = \theta_k - \eta(A\theta_k - B)$

Gradient flow $\frac{d\theta(t)}{dt} = -A\theta + B$



Grad flow converging but to an undesirable value



Requirements for dynamical systems

What type of properties are we interested in?

- Invariance (as in the case of automata)
- State remains bounded
- **State converges to target**
- Bounded input or disturbance gives bounded output (BIBO)

Requirements: Stability

- We will focus on time invariant autonomous systems (closed systems, systems without inputs)
- $\dot{x}(t) = f(x(t)), x_0 \in \mathbb{R}^n, t_0 = 0$ -(1)
- $x(t)$ is the solution
- $|x(t)|$ norm
- $x^* \in \mathbb{R}^n$ is an **equilibrium point** if $f(x^*) = 0$.
- For analysis we will assume $x^* = \mathbf{0}$ to be an equilibrium point of (1) with out loss of generality

Example: Pendulum

Pendulum equation

$$x_1 = \theta \quad x_2 = \dot{\theta}$$

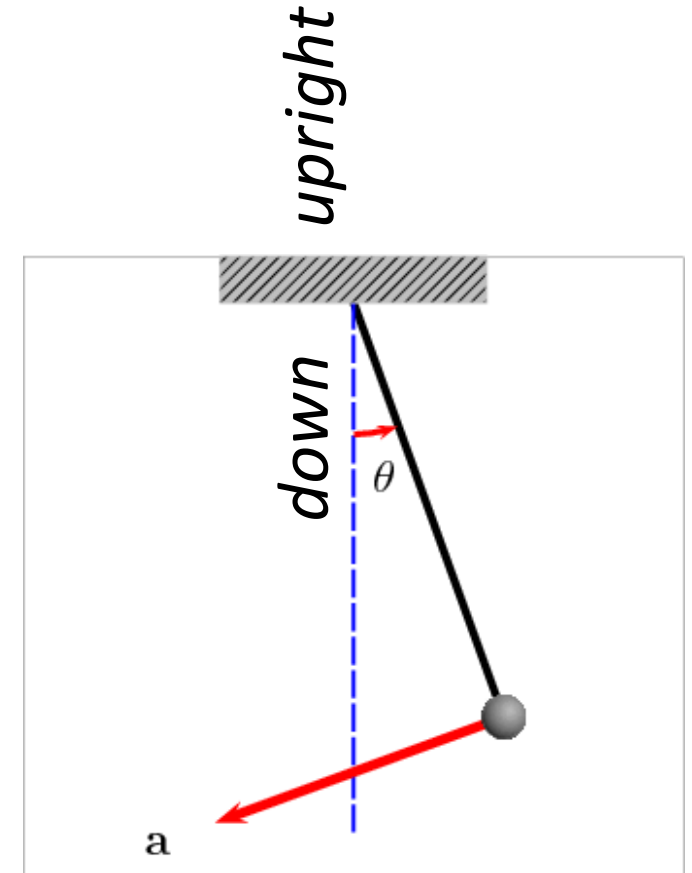
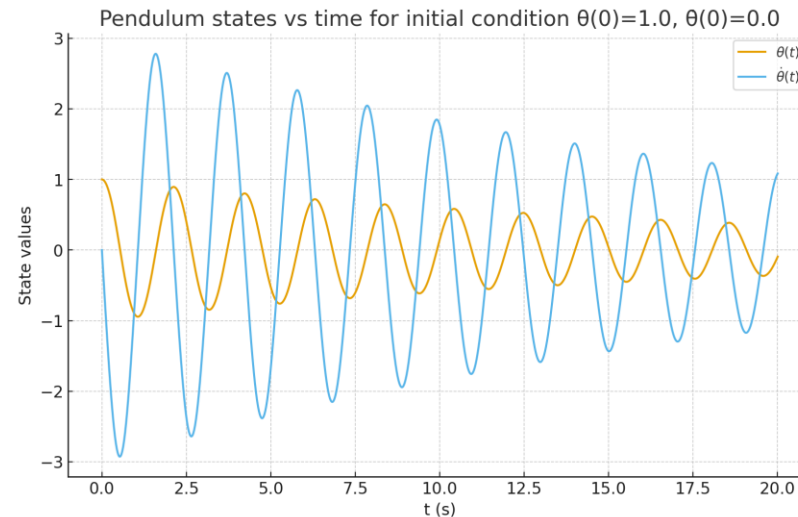
$$x_2 = \dot{x}_1$$

$$\dot{x}_2 = -\frac{g}{l} \sin(x_1) - \frac{k}{m} x_2$$

$$\begin{bmatrix} \dot{x}_2 \\ \dot{x}_1 \end{bmatrix} = \begin{bmatrix} -\frac{g}{l} \sin(x_1) - \frac{k}{m} x_2 \\ x_2 \end{bmatrix}$$

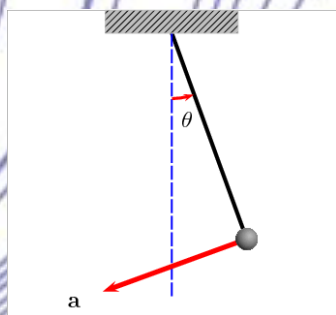
k : friction coefficient

Two equilibrium points: $(0,0)$, $(\pi, 0)$



CW

speed=0



CCW

stable

down

upright

unstable

x (m)

Aleksandr M. Lyapunov

Aleksandr Mikhailovich Lyapunov (June 6 1857–November 3, 1918), Russian mathematician and physicist.



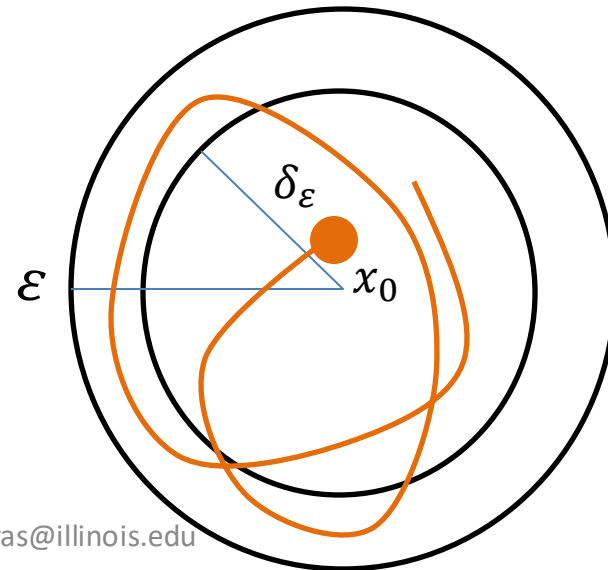
Lyapunov stability

Lyapunov stability: The system (1) is said to be ***Lyapunov stable*** (at the origin) if

$\forall \varepsilon > 0 \exists \delta_\varepsilon > 0$ such that

$$|x_0| \leq \delta_\varepsilon \Rightarrow \forall t \geq 0, |\xi(x_0, t)| \leq \varepsilon.$$

How is this related to
invariants and
reachable states ?

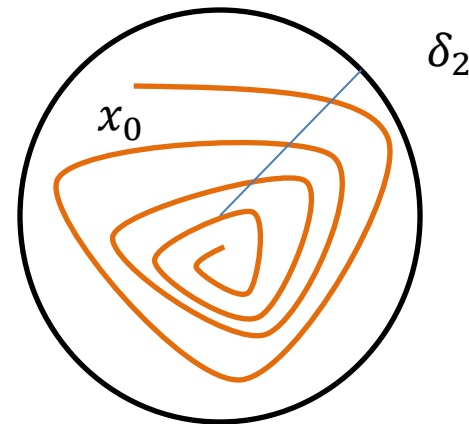


Asymptotically stability

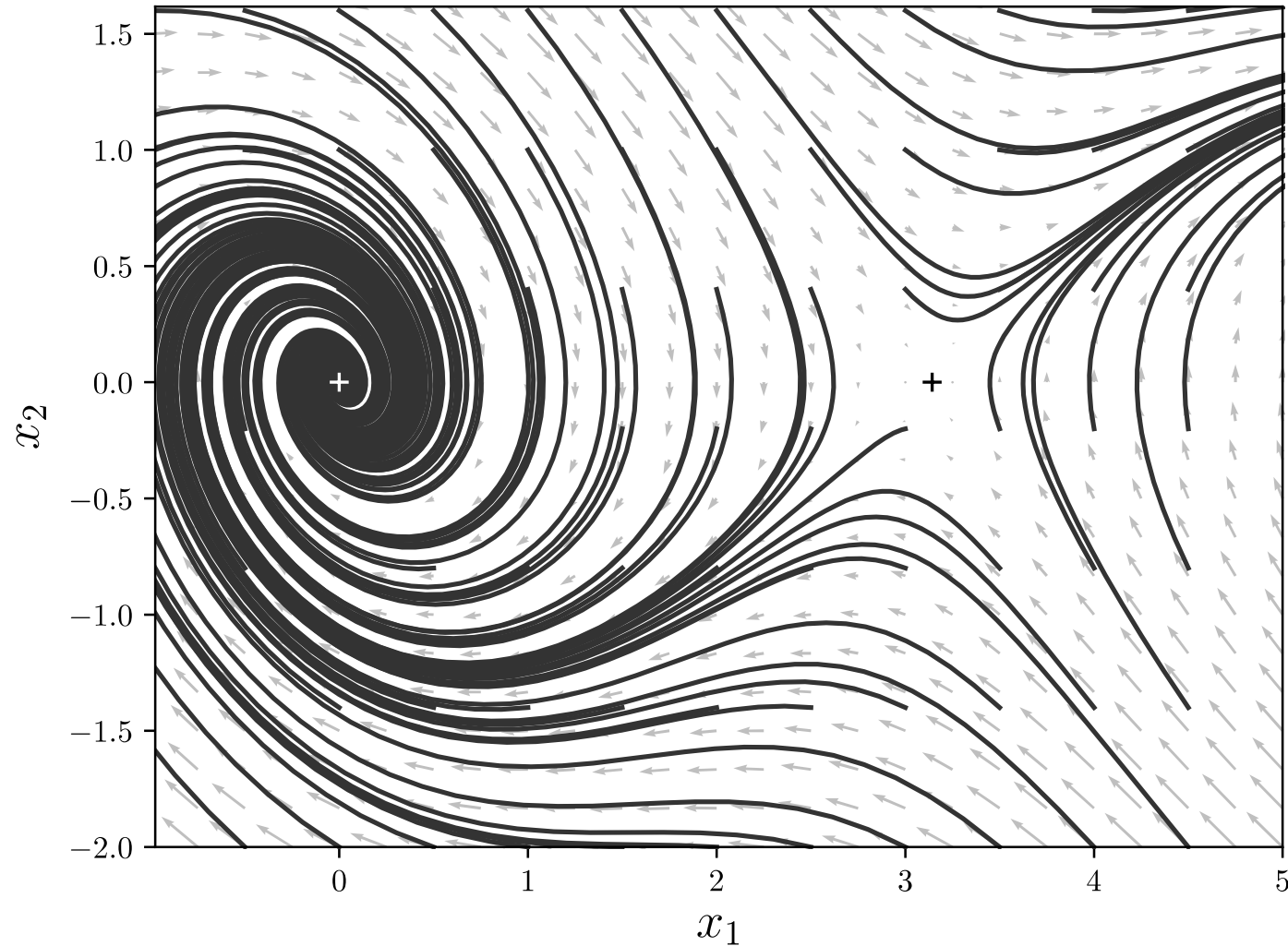
The system (1) is said to be ***Asymptotically stable (at the origin)*** if it is Lyapunov stable and

$\exists \delta_2 > 0$ such that $\forall |x_0| \leq \delta_2$ as $t \rightarrow \infty, |\xi(x_0, t)| \rightarrow \mathbf{0}$.

If the property holds for any δ_2 then **Globally Asymptotically Stable**



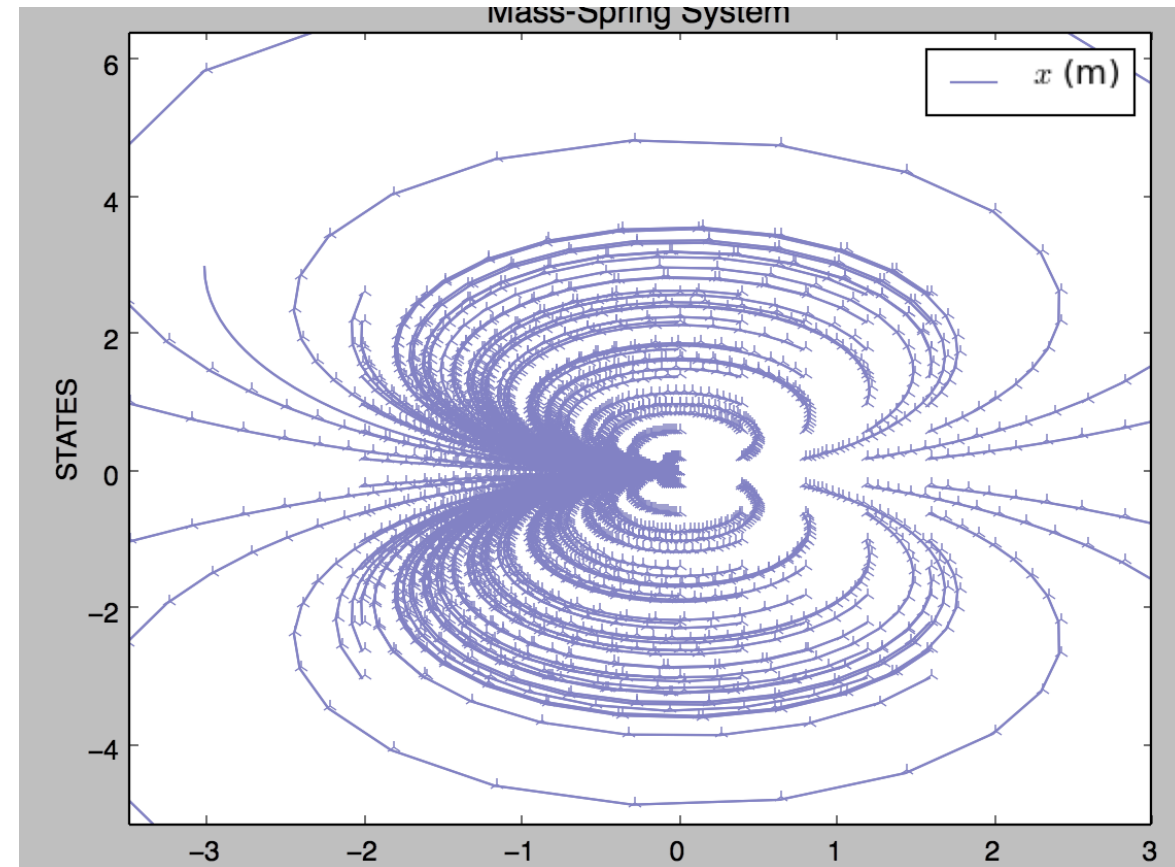
Phase portrait of pendulum with friction



Butterfly*

$$\begin{bmatrix} \dot{x}_2 \\ \dot{x}_1 \end{bmatrix} = \begin{bmatrix} 2x_1x_2 \\ x_1^2 - x_2^2 \end{bmatrix}$$

All solutions converge to 0
but the equilibrium point
(0,0) is not Lyapunov stable



*Not discussed in class

Van der pol oscillator

Van der pol oscillator

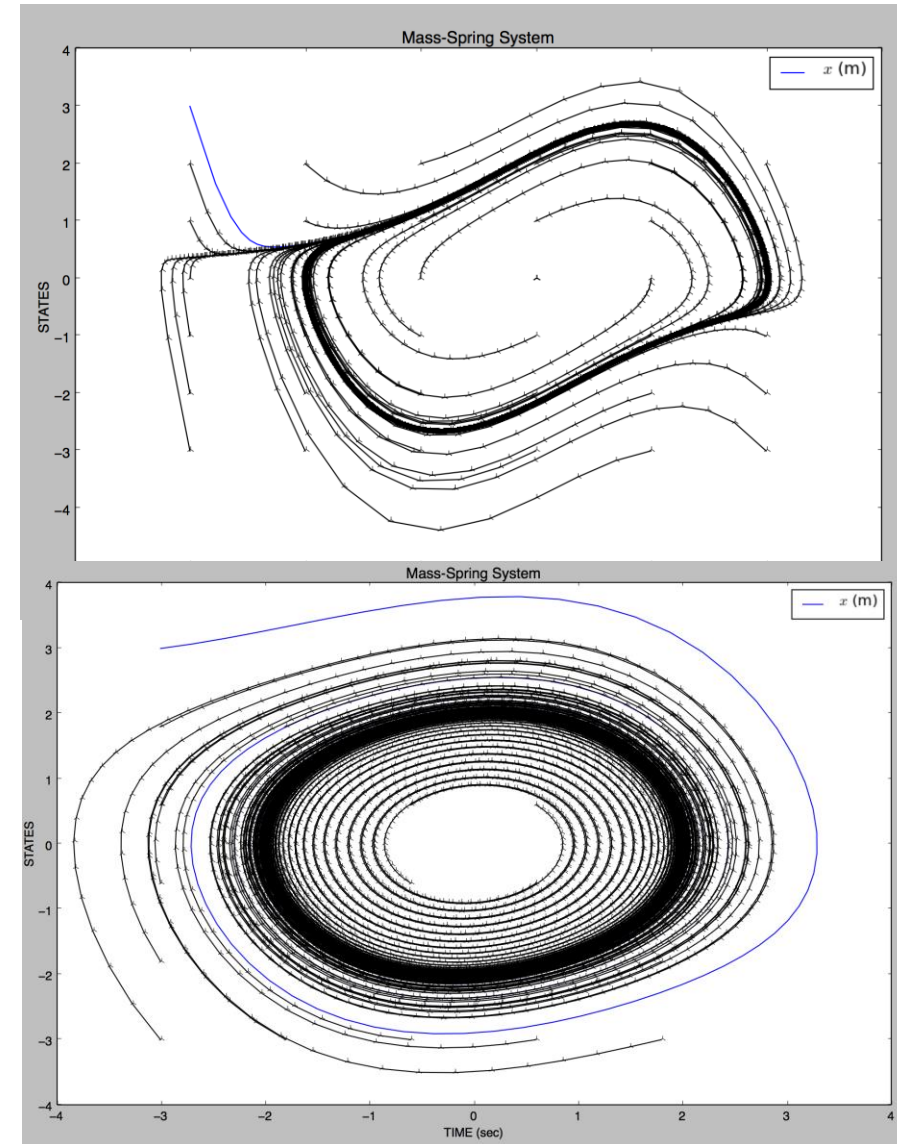
$$\frac{dx^2}{dt^2} - \mu(1 - x^2) \frac{dx}{dt} + x = 0$$

$$x_1 = x; x_2 = \dot{x}_1;$$

coupling coefficient $\mu = 2 \ 0.1$

$$\begin{bmatrix} \dot{x}_2 \\ \dot{x}_1 \end{bmatrix} = \begin{bmatrix} \mu(1 - x_1^2)x_2 - x_1 \\ x_2 \end{bmatrix}$$

stable ?



Stability of solutions* (instead of points)

- For any $\xi \in PC(\mathbb{R}^{\geq 0}, \mathbb{R}^n)$ define the s-norm $\|\xi\|_s = \sup_{t \in \mathbb{R}} \|\xi(t)\|$
- A dynamical system can be seen as an operator that maps initial states to signals $T: \mathbb{R}^n \rightarrow PC(\mathbb{R}^{\geq 0}, \mathbb{R}^n)$
- Lyapunov stability required that this operator is continuous
- The solution ξ^* is **Lyapunov stable** if T is continuous as $\xi^*(0)$. i. e., for every $\varepsilon > 0$ there exists $\delta_\varepsilon > 0$ such that for every $x_0 \in \mathbb{R}^n$ if $|\xi^*(0) - x_0| \leq \delta_\varepsilon$ then $\|T(\xi^*(t)) - T(x_0)\|_s \leq \varepsilon$.

*Not discussed in class

Verifying Termination

How to verify that a loop terminates?

While ($x > 0$)

$x = f(x)$, where $f: \mathbb{Z}^n \rightarrow \mathbb{Z}^n$

Come up with a **ranking function**

$V: \mathbb{Z}^n \rightarrow \mathbb{N}$ such that $V(f(x)) < V(x)$

Input $x_1 \geq x_2 > 0$

While ($x_2 > 0$)

$x_1, x_2 = x_2, x_1 \bmod x_2$

Claim. $V(x_1, x_2) = x_1 + x_2$ works

$V(x_1, x_2) \in \mathbb{N}$

$V(f(x_1, x_2))$

$= x_2 + x_1 \bmod x_2$

$< x_2 + x_2$ [$x_1 \bmod x_2 < x_2$ as loop stops $x_2 = 0$]

$< 2x_2 \leq x_1 + x_2 = V(x_1, x_2)$

Last inequality uses the invariant: $x_1 \geq x_2$

Stability of nonlinear systems

- For any **positive definite** function of state $V: \mathbb{R}^n \rightarrow \mathbb{R}$
 - $V(x) \geq 0$ and $V(x) = 0$ iff $x = 0$
- **Sublevel sets** of $L_p = \{x \in \mathbb{R}^n \mid V(x) \leq p\}$
- $V(x(t))$

V differentiable with continuous first derivative

- $\dot{V} = \frac{dV(x(t))}{dt} = ?$
- $\frac{\partial V}{\partial x} \cdot \frac{d}{dt}(x(t)) = \frac{\partial V}{\partial x} \cdot f(x)$ is also continuous
- V is **radially unbounded** if $\|x\| \rightarrow \infty \Rightarrow V(x) \rightarrow \infty$

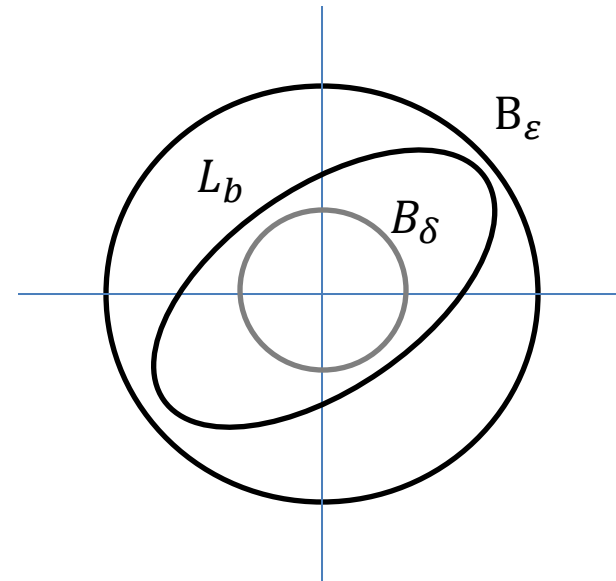
Verifying Stability

Theorem. (Lyapunov) Consider the system (1) with state space $x(t) \in \mathbb{R}^n$ and suppose there exists a **positive definite, continuously differentiable** function $V: \mathbb{R}^n \rightarrow \mathbb{R}$. The system is:

1. Lyapunov stable if $\dot{V}(x(t)) = \frac{\partial V}{\partial x} f(x) \leq 0$, for all $x \neq 0$
2. Asymptotically stable if $\dot{V}(x(t)) < 0$, for all $x \neq 0$
3. It is globally AS if V is also radially unbounded.

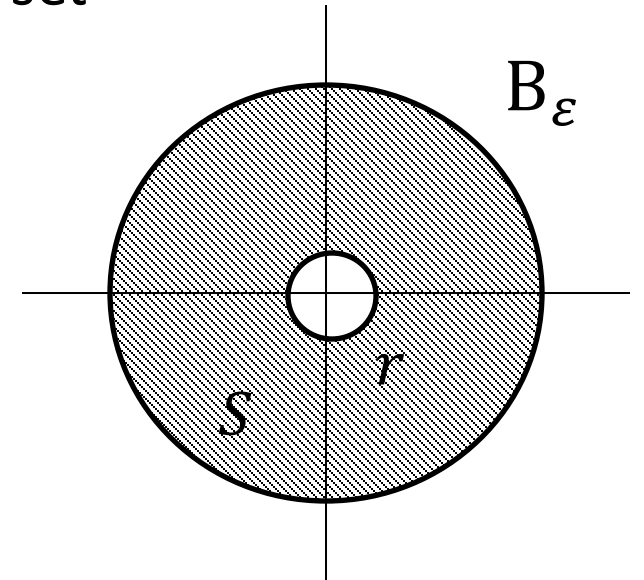
Proof : Lyapunov stable if $\dot{V} \leq 0$

- Assume $\dot{V} \leq 0$
- Fix a ball B_ε around the origin of radius $\varepsilon > 0$.
- Pick a positive number $b < \min_{|x|=\varepsilon} V(x)$.
- Let δ be a radius of ball around origin which is inside $L_b = \{x \mid V(x) \leq b\}$
- $L_b = \{x \mid V(x) \leq b\}$ is an invariant
 - Starting from any $x_0 \in B_\delta$ the trajectory $x(t) \in L_b$. Why? $V(x(t)) \leq V(x_0) \leq b$
 - Therefore $x(t) \in L_b$
- Since $L_b \subseteq B_\varepsilon$ therefore $x(t) \in B_\varepsilon$



Proof sketch: Asymptotically stable if $\dot{V}(x(t)) < 0$

- Assume $\dot{V} < 0$
- Take arbitrary initial state $|x(0)| \leq \delta$, where this δ comes from some ε for Lyapunov stability
- Since $V(x(.)) > 0$ and decreasing along x it has a limit $c \geq 0$ at $t \rightarrow \infty$
- It suffices to show that this limit is actually 0
- Suppose not, $c > 0$ then the solution $x(t)$ evolves in the compact set $S = \{x \mid r \leq |x| \leq \varepsilon\}$ for some sufficiently small r
- Let $d = \max_{x \in S} \dot{V}(x)$ [slowest rate]
- This number is well-defined and negative
- $\dot{V}(x(t)) \leq d$ for all t
- $V(t) \leq V(0) + dt$
- But then eventually $V(t) < c$



Verifying Stability for Linear Systems

Consider the linear system $\dot{x} = Ax$

Theorem.

1. It is asymptotically stable iff all the eigenvalues of A have **strictly** negative real parts (*Hurwitz*).
2. It is Lyapunov stable iff all the eigenvalues of A have real parts that are either zero or negative and the *Jordan blocks* corresponding to the eigenvalues with zero real parts are of size 1.

Jordan decomposition

For every $n \times n$ matrix A , there exists a nonsingular $n \times n$ matrix P such that

$$PAP^{-1} = J = \begin{bmatrix} J_1 & 0 & 0 & \dots & 0 \\ 0 & J_2 & 0 & \dots & 0 \\ 0 & 0 & J_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & J_\ell \end{bmatrix}, \quad J_i = \begin{bmatrix} \lambda_i & 1 & 0 & \dots & 0 \\ 0 & \lambda_i & 1 & \dots & 0 \\ 0 & 0 & \lambda_i & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \lambda_i \end{bmatrix}.$$

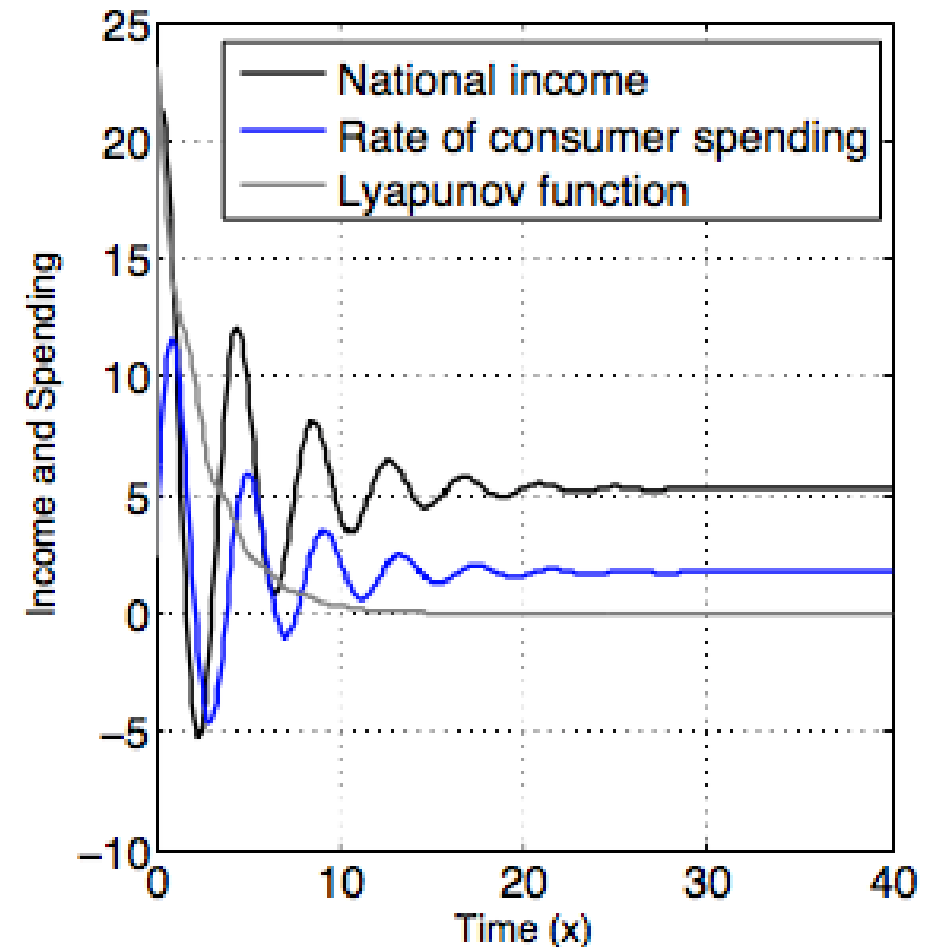
where each J_i is a upper triangular matrix called a Jordan block

Example 1: Simple model of an economy

- x : national income y : rate of consumer spending; g : rate government expenditure
- $\dot{x} = x - \alpha y$
- $\dot{y} = \beta(x - y - g)$
- $g = g_0 + kx$ α, β, k are positive constants
- What is the equilibrium?
- $x^* = \frac{g_0 \alpha}{\alpha - 1 - k\alpha} y^* = \frac{g_0 \alpha}{\alpha - 1 - k\alpha}$
- Dynamics:
- $$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} 1 & -\alpha \\ \beta(1 - k) & -\beta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Example: Simple linear model of an economy

- $\alpha = 3, \beta = 1, k = 0$
- $\lambda_1, \lambda_1^* = (-.25 \pm i 1.714)$
- Negative real parts, therefore, asymptotically stable and the national income and consumer spending rate converge to $x = 1.764$ $y = 5.294$



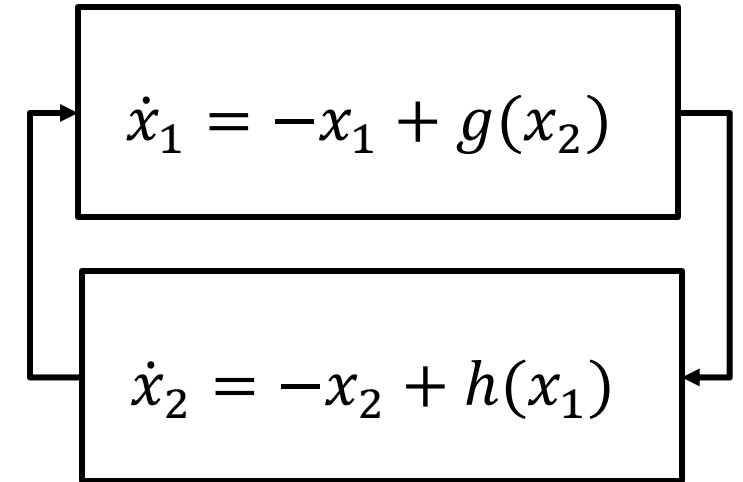
Example 2: Reasoning about stability without solving ODEs

$$\dot{x}_1 = -x_1 + g(x_2); \dot{x}_2 = -x_2 + h(x_1)$$

Given that $|g(x_2)| \leq \frac{|x_2|}{2}, |h(x_1)| \leq \frac{|x_1|}{2}$

- Use $V = \frac{1}{2}(x_1^2 + x_2^2) \geq 0$
- $\dot{V} = x_1 \dot{x}_1 + x_2 \dot{x}_2$
$$= -x_1^2 - x_2^2 + x_1 g(x_2) + x_2 h(x_1)$$
$$\leq -x_1^2 - x_2^2 + \frac{1}{2}(|x_1 x_2| + |x_2 x_1|)$$
$$\leq -\frac{1}{2}(x_1^2 + x_2^2) = -V$$

We conclude global asymptotic stability (in fact global exponential stability) without knowing solutions



$$\begin{aligned} (|x_1| - |x_2|)^2 &\geq 0 \\ x_1^2 + x_2^2 &\geq 2|x_1 x_2| \\ |x_1 x_2| &\leq \frac{1}{2}(x_1^2 + x_2^2) \end{aligned}$$

Proposition. If V is a Lyapunov function then every sublevel set of V is an invariant

Proof. $V(\xi(t)) =$

$$\begin{aligned} &= V(\xi(0)) + \int_0^t \dot{V}(\xi(\tau)) d\tau \\ &\leq V(\xi(0)) \end{aligned}$$

An aside: Checking inductive invariants

- $A = \langle X, Q_0, T \rangle$
 - X : set of variables
 - $Q_0 \subseteq \text{val}(X)$
 - $T \subseteq \text{val}(X) \times \text{val}(X)$ written as a program $x' \subseteq T(x)$
- How do we check that $I \subseteq \text{val}(X)$ is an inductive invariant?
 - $Q_0 \Rightarrow I(X)$
 - $I(X) \Rightarrow I(T(X))$
- Implies that $\text{Reach}_A(Q_0) \subseteq I$ without computing the executions or reachable states of A
- The key is to find such I

Finding Lyapunov Functions

- The key to using Lyapunov theory is to *find* a Lyapunov function and verify that it has the properties
- In general, for nonlinear systems this is hard
- There are several approaches
 - Quadratic Lyapunov functions for linear systems
 - Decide the form/template of the function (e.g., quadratic, polynomial), parameterized by some parameters and find values of the parameters so that the conditions hold (Chapter 3 last section)

Linear autonomous systems

- $\dot{x} = Ax, A \in \mathbb{R}^{n \times n}$
- The Lyapunov equation: $A^T P + PA + Q = 0$
where $P, Q \in \mathbb{R}^{n \times n}$ are symmetric

- Interpretation: $V(x) = x^T P x$ then

$$\dot{V}(x) = (Ax)^T P x + x^T P (Ax)$$

$$\begin{aligned} \text{[using chain rule } \frac{\partial u^T P v}{\partial t} &= \frac{\partial u}{\partial t} P v + \frac{\partial v}{\partial t} P^T u] \\ &= x^T (A^T P + PA) x = -x^T Q x \end{aligned}$$

- If $x^T P x$ is the generalized energy then $-x^T Q x$ is the associated dissipation

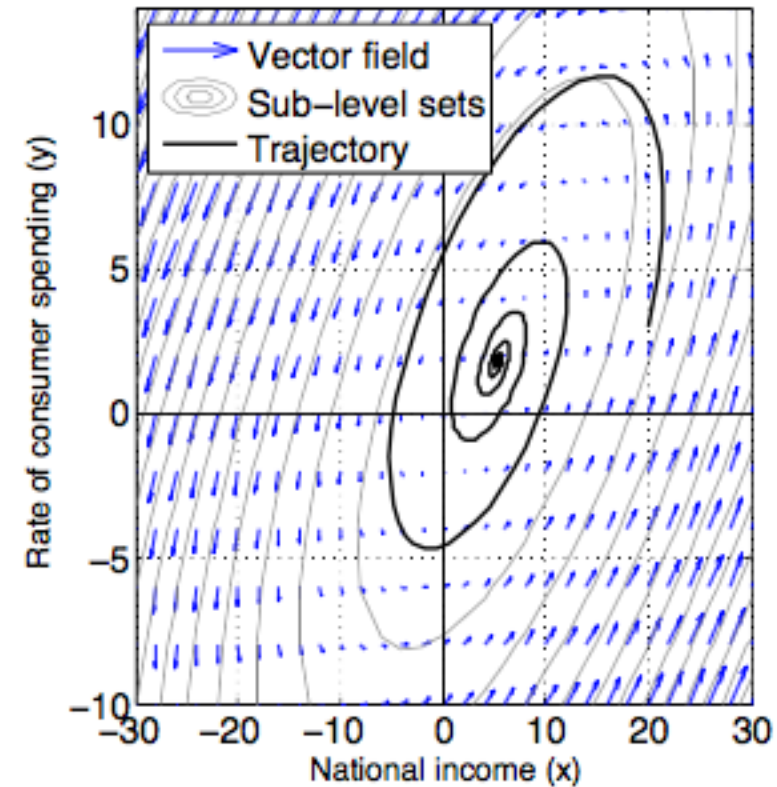
Quadratic Lyapunov Functions

- If $P > 0$ (positive definite)
- $V(x) = x^T P x = 0 \Leftrightarrow x = 0$
- The sub-level sets are ellipsoids
- If $Q > 0$ then the system is globally asymptotically stable

Same example

Lyapunov equations are solved as a set of $\frac{n(n+1)}{2}$ equations in $n(n+1)/2$ variables. Cost $O(n^6)$

Choose $Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ solving Lyapunov equations we get $P = \begin{bmatrix} 2.59 & -2.29 \\ -2.29 & 4.92 \end{bmatrix}$ and we get the quadratic Lyapunov function $(x - x^*)P(x - x^*)^T$ and a sequence of invariants



Converse Lyapunov

Converse Lyapunov theorems show that conditions of the previous theorem are also necessary. For example, if the system is asymptotically stable then there exists a positive definite, continuously differentiable function V , that satisfies the inequalities.

For example if the LTI system $\dot{x} = Ax$ is globally asymptotically stable then there is a quadratic Lyapunov function that proves it.

Small puzzle

- Platonic solids. Solid bodies whose faces are regular polygons, all identical, that meet in identical fashion at every vertex. How many such are there? Exactly five!



Platonic solids.

Wilczek, Frank. A Beautiful Question: Finding Nature's Deep Design (p. 39).